# Grade 10 Physical Sciences: Motion in one dimension

**Physical Sciences · Grade 10 · 50 marks · 120 minutes**

Learner name: _____ Date: _____

Time: 120 minutes Total: 50 marks

## Instructions

- Answer all five questions.
- Show calculations where requested.
- Use evidence from the printed stimulus when a question refers to it.

## Assessment

Read the stimulus and each question carefully.

> *A trolley moves along a straight east-west track; take east as positive. During its first 6 s, it moves east from x = -4 m to x = +20 m in 4 s, then reverses and moves west to x = +8 m at t = 6 s. Assume constant velocity on each segment. In a separate later interval, its velocity changes from +3 m·s⁻¹ to +15 m·s⁻¹ in 4 s.*

**1.1** Distinguish distance and displacement, then calculate both for the trolley's first 6 s of motion. **[8]**

**1.2** For the first 6 s, calculate the trolley's average velocity and average speed. State one difference between them. **[9]**

**1.3** Calculate acceleration during the later interval and state what its sign means. **[10]**

**1.4** Assume the later acceleration stays constant at the value from Question 1.3 for a further 3 s, starting at +15 m·s⁻¹. Calculate the final velocity and signed displacement. Draw a velocity-time graph for this interval or show the trapezium area calculation, and state the direction. **[11]**

**1.5** Compare a position-time graph and a velocity-time graph. For each graph, name both axes and give their units; state what its gradient represents and what a horizontal line means. For the velocity-time graph, state what the signed area represents. **[12]**

Source basis: DBE CAPS Physical Sciences · fet_grade_10_12. See the grade receipt for the official PDF URL and SHA-256.

DRAFT — Qualified educator review required before classroom use.